

When gold disconnects: the size bound, Dirichlet, and the sinks

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Abstract

The hire graph of the 3-free door is built from two inputs. The first is a *size bound*: if an odd prime q divides the door of a prime p , then $p \geq 2q - 1$, so every gold arc runs from a larger prime to a smaller one. The second is *Dirichlet's theorem*, applied to two residue classes, which makes every owner set infinite. From the size bound alone, every gold component of a finite window contains a 2-power door (a Mersenne or Fermat prime), so connectivity of gold is exactly the question of whether the 2-power doors present are joined. From Dirichlet, any two of them are joined in some finite window; the size bound shows how late that can be. Consequently gold is disconnected for arbitrarily large integer windows if and only if there are infinitely many Mersenne or Fermat primes, and each 2-power door $M \geq 7$ forces a whole *octave* $[\omega(M), 2\omega(M) - 2]$ of disconnected windows, not merely one. For M_{31} the picture is exact: assuming the machine-checked connectivity at $X^* = 92\,274\,421$ of the hire-graph note, gold is disconnected for $X^* \leq X \leq 2M_{61} - 2$ precisely when $98\,784\,247\,763 \leq X \leq 627\,065\,224\,920$.

1 Two inputs

The structural results in this note follow from two facts about the door $m_0(p) = p + \chi_3(p)$.

The size bound. The door of p is an even number at most $p + 1$. An odd prime q dividing it therefore divides half of it, so $q \leq (p + 1)/2 < p$. Read from the other side: an owner of q is at least $2q - 1$. This single inequality (Lemma 3.1) is used in four ways: it makes gold acyclic, it caps the primes in a window by $(X + 1)/2$, it gives *door closure* (Corollary 3.3), and it makes every downward gold path terminate. The last use is Theorem 3.5: components of gold are governed by the primes with power-of-two doors.

Dirichlet. Whether $Q \mid m_0(p)$ depends only on p modulo $6Q$, and it selects exactly two invertible classes (Lemma 4.1). Dirichlet's theorem then gives infinitely many owners of any Q coprime to 6, and applying it twice (a common owner, then a hirer of that owner) joins any two primes in some finite window (Theorem 4.3). Dirichlet is qualitative. The size bound is what turns “eventually” into quantitative lower bounds, and Linnik's theorem supplies a general upper bound for the two-step bridge construction; the exact example later in the note uses finite computation.

Results.

- *Sink reduction* (Theorem 3.5): for every window $X \geq 11$, each gold component contains a 2-power door, and gold is connected iff the 2-power doors present are pairwise joined.
- *Isolation octaves* (Theorem 5.1): if $M \geq 7$ is a 2-power door with first owner $\omega(M)$, then gold is disconnected on all of $[\omega(M), 2\omega(M) - 2]$.
- *Equivalence* (Theorem 6.2): gold is disconnected for arbitrarily large integer X iff there are infinitely many Mersenne or Fermat primes.

- M_{31} , *exactly* (Theorem 7.1): the complete disconnected stretch, its end point $X_1 = 627\,065\,224\,921$, and why the least common owner of 5 and M_{31} does not give the first reconnection.

The infinitude of Mersenne primes is open, so item (1) of Theorem 6.2 is not settled unconditionally.

2 Windows, owners and sinks

We use the doors and vertex sets of the hire-graph note [1], but write Γ_X for gold to distinguish it from the hire graph. Throughout, window cutoffs X are integers with $X \geq 5$, and intervals of windows mean their integer points. Write $M_k = 2^k - 1$ and $F_n = 2^{2^n} + 1$. For a prime $p \geq 5$ let $\chi_3(p) = +1$ if $p \equiv 1 \pmod{3}$ and -1 if $p \equiv 2 \pmod{3}$, and write $\chi_3(n)$ likewise for any n coprime to 3.

Lemma 2.1 (Door basics). *Let $p \geq 5$ be prime and $m = m_0(p)$.*

(a) $m \in \{p-1, p+1\}$, m is even, $3 \nmid m$, and $m \geq 4$.

(b) (Inversion) $p = m + \chi_3(m)$.

Proof. (a) If $p \equiv 1 \pmod{3}$ then $m = p+1 \equiv 2 \pmod{3}$; if $p \equiv 2 \pmod{3}$ then $m = p-1 \equiv 1 \pmod{3}$. Evenness and $m \geq 4$ are clear. (b) If $p \equiv 1 \pmod{3}$ then $m = p+1 \equiv 2$, so $\chi_3(m) = -1$ and $m + \chi_3(m) = p$. If $p \equiv 2 \pmod{3}$ then $m = p-1 \equiv 1$, so $\chi_3(m) = +1$ and $m + \chi_3(m) = p$. $\square$

We say p *owns* an odd prime $q \neq 3$ if $q \mid m_0(p)$; such q is automatically ≥ 5 . For an integer $X \geq 5$ put $P_X = \{p \text{ prime} : 5 \leq p \leq X\}$ and

$$S_X = \{2\} \cup \{q \text{ odd prime} : q \mid m_0(p) \text{ for some } p \in P_X\}.$$

The *gold graph* Γ_X has vertex set $S_X \setminus \{2\}$ and a directed arc $p \rightarrow q$ whenever $q \mid m_0(p)$; the undirected gold graph forgets orientation. Connectivity always refers to the undirected gold graph. (The hire graph proper is the cone over this graph with apex 2, which is why finite windows are connected whether or not gold is; see [1].)

More generally, for an odd integer $Q > 1$ with $3 \nmid Q$, an owner of Q is a prime $p \geq 5$ such that $Q \mid m_0(p)$. Let $\omega(Q)$ denote its least owner, with $\omega(Q) = \infty$ if none exists (Corollary 4.2 shows this never happens). For an odd prime $q \neq 3$, by definition

$$q \in S_X \iff \omega(q) \leq X. \tag{1}$$

So $\omega(q)$ is the *hire time* of q : the first window in which q is a vertex. In this notation the “first owner r_0 of M ” of the earlier version is $\omega(M)$; we keep r_0 for $\omega(M_{31})$.

Since arcs between two vertices of S_X are determined by the doors alone, we have monotonicity:

$$X \leq X' \implies S_X \subseteq S_{X'} \text{ and } \Gamma_X \text{ is an induced subgraph of } \Gamma_{X'}. \tag{2}$$

In particular two vertices joined in Γ_X stay joined in every larger window. Connectivity of the *whole* graph is not monotone: a new vertex can enter with no arcs.

An odd prime $M \neq 3$ is a *2-power door* if $m_0(M) = 2^k$. Write $\mathcal{M}$ for the set of these. By the hire-graph note (Lemma 10 there), $\mathcal{M}$ consists of the Mersenne primes $2^p - 1$ (p odd) and the Fermat primes $2^{2^n} + 1$, $n \geq 1$:

$$\mathcal{M} = \{5, 7, 17, 31, 127, 257, 8191, 65537, 131071, 524287, 2^{31} - 1, \dots\}.$$

A *sink* of Γ_X is a vertex with no outgoing arc. Note that $5 \in \mathcal{M}$, and that $\omega(5) = 11$: the doors of 5, 7 are 4, 8, and $m_0(11) = 10$. For $5 \leq X < 11$, gold has no vertices; for $X = 11, 12$, it is the connected singleton $\{5\}$. At $X = 13$, the sink 7 enters and gold becomes disconnected. The value $X^* = 92\,274\,421$ used below is a later connected window reported in [1]; the proof needs its connectivity, not a claim that it is the first connected window.

3 The size bound

Lemma 3.1 (Size bound). *Let p, q be odd primes, $p \geq 5$, $q \neq 3$, with $q \mid m_0(p)$. Then*

$$2q \leq m_0(p) \leq p + 1, \quad \text{hence} \quad q \leq \frac{m_0(p)}{2} \leq \frac{p + 1}{2} < p, \quad p \geq 2q - 1.$$

Proof. By Lemma 2.1, $m_0(p)$ is even and at most $p + 1$. Since q is odd and divides an even number, $2q \mid m_0(p)$, so $m_0(p) \geq 2q$. Combine, and note $(p + 1)/2 < p$ for $p > 1$. $\square$

The proof uses only that the door is even and at most $p + 1$. Note the inequality is $q \leq (p + 1)/2$, not $q < (p + 1)/2$: equality occurs (for instance $p = 13$, $q = 7$), The strict consequence $q < p$ gives descent, while the full bound gives the isolation intervals below.

Remark 3.2 (Sharpness). Equality $q = m_0(p)/2$ means $m_0(p) = 2q$. By Lemma 2.1(b) the only prime that can have door $2q$ is $p = 2q + \chi_3(2q) = 2q - \chi_3(q)$, that is $2q - 1$ if $q \equiv 1 \pmod{3}$ and $2q + 1$ if $q \equiv 2 \pmod{3}$. So $\omega(q) \geq 2q - \chi_3(q)$, with equality exactly when that number is prime. The next candidate has door $4q$ and is $4q + \chi_3(q)$. This is the Sophie Germain pattern of the hire-graph note (Lemma 4 there) seen as the extremal case.

Corollary 3.3 (Consequences of the size bound). (a) *Acyclicity. Every gold arc $p \rightarrow q$ has $q < p$; the graph is acyclic and no two vertices are joined by arcs in both directions.*

(b) *Ceiling and floor. If $q \in S_X \setminus \{2\}$ then $q \leq (X + 1)/2$. Equivalently $\omega(q) \geq 2q - 1$ for every odd prime $q \neq 3$.*

(c) *Door closure. If $q \in S_X \setminus \{2\}$ and r is an odd prime dividing $m_0(q)$, then $r \in S_X$.*

(d) *Chains of owners. If w owns q then $\omega(w) \geq 2w - 1 \geq 2\omega(q) - 1$.*

Proof. (a) is the inequality $q < p$. (b) Let $q \in S_X$, so $q \mid m_0(p)$ for some $p \leq X$; then $q \leq (p + 1)/2 \leq (X + 1)/2$. The equivalent form is (1). (c) Such q is a prime ≥ 5 with $q \leq (X + 1)/2 \leq X$, so $q \in P_X$; every odd prime factor of its door is then in S_X by definition. (d) $\omega(w) \geq 2w - 1$ by (b), and $w \geq \omega(q)$ because w is an owner of q . $\square$

Parts (b) and (d) are the quantitative half of the note: a vertex q cannot be hired before $2q - 1$, and a vertex hired through an owner w is hired no earlier than $2w - 1$, so the corresponding size lower bounds grow by the recurrence $a \mapsto 2a - 1$.

Proposition 3.4 (Descent). *Let $v \in S_X \setminus \{2\}$. There is a strictly decreasing gold path $v = v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_n$ in Γ_X ending at a vertex v_n with $m_0(v_n)$ a power of 2; that is, $v_n \in \mathcal{M}$.*

Proof. If $m_0(v_i)$ has an odd prime factor r , then $r \in S_X$ by Corollary 3.3(c), so $v_i \rightarrow r$ is an arc of Γ_X , and $r < v_i$ by (a). A strictly decreasing sequence of primes terminates; at the terminal vertex no odd prime divides the door. $\square$

Theorem 3.5 (Sink reduction). *Let $X \geq 11$ and put $D_X = \{M \in \mathcal{M} : \omega(M) \leq X\}$.*

(i) *D_X is the set of sinks of Γ_X , and $5 \in D_X$.*

- (ii) Every vertex of Γ_X is joined to some element of D_X by a decreasing gold path.
- (iii) Every connected component of Γ_X contains an element of D_X . Consequently the number of components of Γ_X equals the number of classes of D_X under connectivity, and Γ_X is connected if and only if the elements of D_X lie in one component.

Proof. (i) By (1), $M \in S_X$ iff $\omega(M) \leq X$. A vertex q has no outgoing arc iff no odd prime dividing $m_0(q)$ lies in S_X , iff (by closure) $m_0(q)$ has no odd prime factor, iff $q \in \mathcal{M}$. Since $\omega(5) = 11 \leq X$, $5 \in D_X$. (ii) is Proposition 3.4. (iii) follows from (ii); the count is then immediate. $\square$

The theorem says: the only obstruction to connectivity is between sinks. Generic vertices are always attached, through their descent, to a sink.

Remark 3.6 (Check). I replayed every window $X \leq 10^7$ (all 664 577 primes) with union-find and compared the component count of Γ_X with the number of classes of D_X : they agree at every window (`hire_checks.py`, part A). This is a test of Theorem 3.5, whose proof above is complete, and it also reproduces the component counts of the hire-graph note at $X = 30, 50, 100, 200, 10^4, 5 \cdot 10^5, 10^7$, namely 2, 2, 4, 4, 2, 3, 3.

4 Dirichlet

Fix an odd integer $Q \geq 3$ with $3 \nmid Q$, and put $s = \chi_3(Q)$, so $Q \equiv s \pmod{3}$.

Lemma 4.1 (Two classes). *For a prime $p \geq 5$,*

$$Q \mid m_0(p) \iff p \equiv 2Q - s \text{ or } p \equiv 4Q + s \pmod{6Q}.$$

Moreover the owners of Q are exactly the primes among the numbers

$$p_j(Q) = 2jQ - \chi_3(j)s, \quad j \geq 1, \ 3 \nmid j, \tag{3}$$

where $m_0(p_j) = 2jQ$; the first class corresponds to $j \equiv 1 \pmod{3}$ and the second to $j \equiv 2 \pmod{3}$.

Proof. Suppose $Q \mid m_0(p) =: m$. Then $2Q \mid m$ (m even, Q odd), say $m = 2jQ$, and $3 \nmid j$ because $3 \nmid m$. As $\chi_3(2) = -1$, $\chi_3(m) = -\chi_3(j)s$, and inversion (Lemma 2.1(b)) gives $p = 2jQ - \chi_3(j)s$. Writing $j = 1 + 3i$ or $j = 2 + 3i$ yields $p \equiv 2Q - s$ or $4Q + s \pmod{6Q}$. Conversely let $p \equiv 2Q - s \pmod{6Q}$. Modulo 3, $p \equiv 2s - s = s$, so $\chi_3(p) = s$ and $m_0(p) = p + s \equiv (2Q - s) + s \equiv 0 \pmod{Q}$. If $p \equiv 4Q + s$, then modulo 3, $p \equiv s + s \equiv -s$, so $\chi_3(p) = -s$ and $m_0(p) = p - s \equiv 4Q \equiv 0 \pmod{Q}$. $\square$

Corollary 4.2 (Infinitely many owners). *Both classes in Lemma 4.1 are coprime to $6Q$: the numbers $2Q - s$ and $4Q + s$ are odd, are $\equiv s$ and $\equiv -s \pmod{3}$ respectively (so nonzero), and are $\equiv -s$ and $\equiv s \pmod{Q}$ respectively (so coprime to Q). By Dirichlet's theorem [2], each class contains infinitely many primes. Hence every odd Q coprime to 3 has infinitely many owners. In particular $\omega(Q) < \infty$ for every such Q , every odd prime $\neq 3$ is eventually a vertex, and $\omega(Q) \geq 2Q - 1$, since an even door divisible by the odd integer Q is at least $2Q$.*

The parametrisation (3) is also the search procedure: $\omega(Q)$ is the least prime among $p_1, p_2, p_4, p_5, p_7, \dots$. For example $\omega(M_{31}) = r_0 = 98\,784\,247\,763 = 46M_{31} + 1$ is $p_{23}(M_{31})$, since $\chi_3(23M_{31}) = -1$.

Theorem 4.3 (Bridging). *Let q_1, q_2 be odd primes different from 3. There is a window X such that q_1, q_2 lie in the same component of $\Gamma_{X'}$ for every $X' \geq X$.*

Proof. Put $Q = q_1 q_2$ (or $Q = q_1$ if $q_1 = q_2$). Dirichlet step 1: by Corollary 4.2 choose a prime r with $Q \mid m_0(r)$; then $r \geq 5$ and r owns both q_i . Dirichlet step 2: choose a prime u with $r \mid m_0(u)$ and set $X = u$. Then $r \in S_X$. By door closure, $q_1, q_2 \in S_X$, and the arcs $r \rightarrow q_1$ and $r \rightarrow q_2$ are arcs of Γ_X ; so q_1, q_2 are joined through r . Persistence is (2). $\square$

Taking $q_1 = 5$ recovers the theorem of the hire-graph note that every odd prime other than 3 lies in the component of 5 in the infinite gold graph $\Gamma_\infty = \bigcup_X \Gamma_X$. (By Corollary 4.2, Γ_∞ has every odd prime $\neq 3$ as a vertex.)

Proposition 4.4 (Bounds for a two-step bridge). *Let q be an odd prime, $q \neq 3, 5$.*

(a) *If a prime r owns both 5 and q , and a prime u owns r , then*

$$r \geq 10q - 1, \quad u \geq 2r - 1 \geq 20q - 3.$$

Thus no common-owner bridge of this type can be live below $20q - 3$.

(b) *For any fixed admissible exponent L in Linnik's theorem, one can choose such primes r, u with*

$$r \ll q^L, \quad u \ll r^L \ll q^{L^2},$$

where the implied constants are independent of q .

Proof. (a) The door $m_0(r)$ is an even multiple of $5q$, so $m_0(r) \geq 10q$ and $r \geq 10q - 1$. Apply Lemma 3.1 to u, r . (b) Choose r as the least prime in either class of Lemma 4.1 for $Q = 5q$, and then choose u as the least prime in either class for $Q = r$. Linnik's theorem, applied with moduli $30q$ and $6r$, gives the stated bounds. For example, $L = 5.18$ is supplied by [3, Theorem 1.1]. $\square$

These bounds concern the specified common-owner construction: its earliest live window is at least $20q - 3$ and at most Cq^{L^2} for a constant C . They do not give a lower bound for an arbitrary path from q to 5. Section 7 gives an example where a longer path appears before the least common owner of 5 and q is hired.

5 Isolation octaves

For an odd prime $q \neq 3$, let

$$\alpha(q) = \min\{\omega(w) : w \text{ prime, } q \mid m_0(w)\}$$

be the first window in which some arc *into* q is live: an arc $w \rightarrow q$ needs an owner w of q that is itself a vertex, and $w \in S_X$ iff $\omega(w) \leq X$.

Theorem 5.1 (Isolation octave). *Let $M \in \mathcal{M}$ with $M \geq 7$.*

(a) $\alpha(M) \geq 2\omega(M) - 1$.

(b) *For every X with $\omega(M) \leq X < \alpha(M)$, the prime M is an isolated vertex of Γ_X , and Γ_X is disconnected.*

(c) *In particular gold is disconnected for all X with $\omega(M) \leq X \leq 2\omega(M) - 2$.*

Proof. (a) If w owns M then $\omega(w) \geq 2w - 1 \geq 2\omega(M) - 1$ by Corollary 3.3(d). (b) $M \in S_X$ by (1). It has no outgoing arc, since $m_0(M)$ is a power of 2. An incoming arc $w \rightarrow M$ in Γ_X needs $M \mid m_0(w)$ and $w \in S_X$, that is $\omega(w) \leq X$, which contradicts $X < \alpha(M)$. Finally $\omega(M) \geq 2M - 1 \geq 13 > 11 = \omega(5)$, so $5 \in S_X$ is a second vertex, and $M \neq 5$. (c) follows from (a) and (b). $\square$

The interval contains exactly $\omega(M) - 1 \geq 2M - 2$ integer cutoffs. We call it an isolation octave because its upper endpoint is just below twice its lower endpoint. Table 1 lists the intervals for every known 2-power door up to M_{31} ; for all of them below M_{31} , I verified by brute force that gold is disconnected on every window of the interval (`hire_checks.py`, part A).

M	$\omega(M)$	j	interval $[\omega, 2\omega - 2]$	checked
7	13	1	[13, 24]	yes
17	67	2	[67, 132]	yes
31	61	1	[61, 120]	yes
127	509	2	[509, 1 016]	yes
257	4 111	8	[4 111, 8 220]	yes
8 191	16 381	1	[16 381, 32 760]	yes
65 537	262 147	2	[262 147, 524 292]	yes
131 071	1 834 993	7	[1 834 993, 3 669 984]	yes
524 287	1 048 573	1	[1 048 573, 2 097 144]	yes
$2^{31} - 1$	98 784 247 763	23	[98 784 247 763, 197 568 495 524]	(Thm. 7.1)

Table 1: First owners and guaranteed isolation octaves; $m_0(\omega(M)) = 2jM$.

The numbers 4111, 16381 and 1 048 573 are the first owners of 257, 8191 and 524 287, respectively. Theorem 5.1 keeps each sink isolated throughout its guaranteed octave. A larger satellite containing its owner can appear only later, once that owner is itself hired; the theorem does not assert persistence of any particular multi-vertex satellite.

6 The equivalence

Theorem 6.1 (Connected stretches). *Let $11 \leq X_0 \leq X_1$. If Γ_{X_0} is connected and no $M \in \mathcal{M}$ has $X_0 < \omega(M) \leq X_1$, then Γ_X is connected for every $X_0 \leq X \leq X_1$.*

Proof. For such X we have $D_X = D_{X_0}$. The elements of D_{X_0} lie in one component of Γ_{X_0} , hence of Γ_X by (2). Apply Theorem 3.5(iii). $\square$

Theorem 6.2. *The following are equivalent.*

1. *There are arbitrarily large integer windows X on which undirected gold is disconnected.*
2. *There are infinitely many 2-power-door primes.*
3. *There are infinitely many Mersenne primes, or infinitely many Fermat primes.*

Proof. (2) $\Rightarrow$ (1). For each $M \in \mathcal{M}$ with $M \geq 7$, Theorem 5.1(c) gives disconnected windows starting at $\omega(M) \geq 2M - 1$, which tends to infinity with M .

(1) $\Rightarrow$ (2). Suppose $\mathcal{M}$ is finite. By Theorem 4.3, applied to the finitely many pairs from $\mathcal{M}$ and using (2), there is $X_0 \geq \max_{M \in \mathcal{M}} \omega(M)$ such that all elements of $\mathcal{M}$ lie in one component of Γ_{X_0} . For every $X \geq X_0$ we have $D_X = \mathcal{M}$, and these primes remain in one component by (2). By Theorem 3.5(iii), Γ_X is connected for every $X \geq X_0$, so only finitely many windows are disconnected.

(2) $\Leftrightarrow$ (3) is the classification of 2-power doors as Mersenne primes with odd exponent and Fermat primes greater than 3. $\square$

Fermat primes are believed finite (five are known, the largest 65537), and the infinitude of Mersenne primes is open; so Theorem 6.2 identifies the obstruction rather than settling item (1). The forward construction supplies an interval of disconnected windows for every new sink. The converse uses a single finite window joining all sinks when their set is finite.

Remark 6.3 (Isolation and reconnection). For $M \in \mathcal{M} \setminus \{5\}$, define

$$\beta(M) = \min\{X \geq 11 : M \text{ and } 5 \text{ lie in the same component of } \Gamma_X\}.$$

Theorem 4.3 makes $\beta(M)$ finite, and $\alpha(M) \leq \beta(M)$. At $\alpha(M)$ the sink first acquires an incoming edge; it may still lie in a separate component. The two times coincide for M_{31} , as shown below, but equality is not assumed in general. The Lenstra–Pomerance–Wagstaff heuristic [4] predicts infinitely many Mersenne primes, hence, through Theorem 6.2, arbitrarily large disconnected windows. No density assertion is made here: that would require further control of sink arrivals and their connection times $\beta(M)$, including any contribution from Fermat primes.

7 M_{31} , exactly

Let $M = M_{31} = 2^{31} - 1 \equiv 1 \pmod{3}$, so $s = +1$. By Lemma 4.1 the owners of M are the primes $tM - 1$ with $t \equiv 2 \pmod{6}$ and $tM + 1$ with $t \equiv 4 \pmod{6}$ ($t = 2j$). The least is

$$r_0 = \omega(M_{31}) = 46M + 1 = 98\,784\,247\,763, \quad m_0(r_0) = 2 \cdot 23 \cdot M_{31},$$

and no smaller t gives a prime; the floor candidate $2M - 1 = 4\,294\,967\,293 = 9241 \cdot 464\,773$ is composite. Two searches of the same kind give the hire time of r_0 itself and all owners that could matter. The hire time of r_0 is

$$\omega(r_0) = 14r_0 + 1 = 1\,382\,979\,468\,683,$$

the least prime among $tr_0 + 1$ for $t \equiv 2 \pmod{6}$ and $tr_0 - 1$ for $t \equiv 4 \pmod{6}$, since $r_0 \equiv 2 \pmod{3}$. An owner w of M_{31} can be hired only at $X \geq 2w - 1$ (Corollary 3.3(b)), so the only owners that can be hired at or before $\omega(r_0)$ satisfy $2w - 1 \leq 1\,382\,979\,468\,683$. Exactly thirteen owners satisfy this necessary size bound:

t	owner $w = tM_{31} \mp 1$	door $m_0(w) = tM_{31}$	hire time $\omega(w)$
46	98 784 247 763	$2 \cdot 23 \cdot M_{31}$	1 382 979 468 683
136	292 057 775 993	$2^3 \cdot 17 \cdot M_{31}$	$> \omega(r_0)$
140	300 647 710 579	$2^2 \cdot 5 \cdot 7 \cdot M_{31}$	$> \omega(r_0)$
146	313 532 612 461	$2 \cdot 73 \cdot M_{31}$	627 065 224 921
184	395 136 991 049	$2^3 \cdot 23 \cdot M_{31}$	$> \omega(r_0)$
196	420 906 794 813	$2^2 \cdot 7^2 \cdot M_{31}$	841 813 589 627
220	472 446 402 341	$2^2 \cdot 5 \cdot 11 \cdot M_{31}$	$> \omega(r_0)$
226	485 331 304 223	$2 \cdot 113 \cdot M_{31}$	$> \omega(r_0)$
238	511 101 107 987	$2 \cdot 7 \cdot 17 \cdot M_{31}$	$> \omega(r_0)$
242	519 691 042 573	$2 \cdot 11^2 \cdot M_{31}$	$> \omega(r_0)$
272	584 115 551 983	$2^4 \cdot 17 \cdot M_{31}$	$> \omega(r_0)$
280	601 295 421 161	$2^3 \cdot 5 \cdot 7 \cdot M_{31}$	$> \omega(r_0)$
296	635 655 159 511	$2^3 \cdot 37 \cdot M_{31}$	$> \omega(r_0)$

Table 2: Owners w of M_{31} with $2w - 1 \leq \omega(r_0)$. The entry $> \omega(r_0)$ means no prime owner of w exists up to $\omega(r_0)$.

The minimum is attained by $w = 313\,532\,612\,461 = 146M_{31} - 1$, and

$$X_1 := \alpha(M_{31}) = 627\,065\,224\,921 = 2w - 1,$$

the extremal case of the size bound (Remark 3.2): w is hired by the prime $2w - 1$, whose door is $2w$. The door of w is $2 \cdot 73 \cdot M_{31}$, so at X_1 there are arcs $w \rightarrow M_{31}$ and $w \rightarrow 73$, and 73 lies in the mainland by the chain

$$73 \rightarrow 37 \rightarrow 19 \rightarrow 5, \quad m_0(73) = 2 \cdot 37, \quad m_0(37) = 2 \cdot 19, \quad m_0(19) = 2^2 \cdot 5.$$

Theorem 7.1 (The M_{31} stretch). *Assume that Γ_{X^*} is connected for $X^* = 92\,274\,421$ (hire-graph note, machine-checked, not Lean). Then for every X with $X^* \leq X \leq 2M_{61} - 2 = 2^{62} - 4$, the gold graph Γ_X is disconnected if and only if*

$$98\,784\,247\,763 = \omega(M_{31}) \leq X \leq \alpha(M_{31}) - 1 = 627\,065\,224\,920.$$

Proof. Which sinks occur. If $M \in \mathcal{M}$ is in S_X then $M \leq (X+1)/2 < M_{61}$ when $X \leq 2M_{61} - 2$. For a Mersenne prime below M_{61} , the exponent is a prime less than 61. The exponents 3, 5, 7, 13, 17, 19, 31 give primes; the other odd prime exponents are excluded by the following proper divisors:

p	11	23	29	37	41	43	47	53	59
$d \mid 2^p - 1$	23	47	233	223	13367	431	2351	6361	179951

The Fermat primes below M_{61} , excluding 3, are 5, 17, 257, 65537: $F_5 = 641 \cdot 6700417$ is composite and $F_6 > M_{61}$. So $\mathcal{M} \cap [1, M_{61}) = \{5, 7, 17, 31, 127, 257, 8191, 65537, 131071, 524287, M_{31}\}$, and all except M_{31} have $\omega \leq 1\,834\,993 < X^*$ (Table 1). Hence $D_X = D_{X^*}$ for $X^* \leq X < \omega(M_{31})$, and $D_X = D_{X^*} \cup \{M_{31}\}$ for $\omega(M_{31}) \leq X \leq 2M_{61} - 2$.

Before $\omega(M_{31})$. D_{X^*} lies in one component of Γ_{X^*} (connected), hence of Γ_X by (2); Theorem 6.1 gives connectedness on $[X^*, \omega(M_{31}) - 1]$.

On $[\omega(M_{31}), X_1 - 1]$. By Theorem 5.1(b), M_{31} is isolated and gold is disconnected, provided $\alpha(M_{31}) = X_1$. That holds because every owner w of M_{31} with $\omega(w) < X_1$ would satisfy $2w - 1 \leq \omega(w) < X_1 < \omega(r_0)$, so it appears in Table 2; and there the smallest hire time is X_1 .

From X_1 on. At X_1 the vertex w is present with arcs to M_{31} and to 73; by door closure 37, 19, 5 $\in S_{X_1}$ and the chain above is live, so M_{31} is joined to 5, hence to all of D_{X^*} . Thus D_{X_1} is in one component, and by (2) it stays so on $[X_1, 2M_{61} - 2]$, where no further sink enters; Theorem 3.5 gives connectedness. $\square$

Remark 7.2 (The old bridge owner). The previous version certified $v = 300\,647\,710\,579 = 140M_{31} - 1$, with $m_0(v) = 2^2 \cdot 5 \cdot 7 \cdot M_{31}$, as the least common owner of 5 and M_{31} and noted that a bridge *owner* is not a bridge *vertex*. Table 2 settles this: v has no owner up to $\omega(r_0) = 1\,382\,979\,468\,683$, far later than the actual reconnection $X_1 \approx 6.27 \cdot 10^{11}$. The reconnection occurs through a different owner, one whose door has no factor 5; M_{31} reaches 5 along $73 \rightarrow 37 \rightarrow 19 \rightarrow 5$. It also confirms that the “first direct 5– M_{31} bridge” is not the first connecting vertex.

Remark 7.3 (Cost). The earlier version called a full component check at these scales “past home computing”. The exact answer above needs only Lemma 4.1’s parametrisation (exactly 108 admissible candidates with $t \leq 322$) followed by bounded owner searches for the thirteen primes found. All these searches use primality tests of numbers below $2 \cdot 10^{12}$; see `hire_checks.py`, part B. The reason is Theorem 3.5: connectivity is decided by the sinks, so no graph rebuild is required.

8 Formalisation, reproducibility and changes

Lean. The Lean development (`Hire/GoldDisconnects.lean`) checks the size bound $2q \leq p+1$, door closure, descent to a 2-power door, and the sink reduction: the sinks are the 2-power doors hired in the window, and gold is connected if and only if those sinks lie in one component. It also checks the single-window isolation lemma and the implication from infinitely many 2-power doors to infinitely many disconnected windows. The octave in its interval form, the stretch, Theorem 7.1, and the converse of the equivalence are not formalised. The computed data (Tables 1 and 2, the value X_1) are results of finite deterministic calculations, not Lean theorems or separately supplied primality certificates.

Reproducibility. `hire_checks.py` (Python 3, `sympy`) performs two tasks. Part A replays all windows $X \leq L$ (default 10^6 ; 10^7 was used for the statements above) and checks Theorem 3.5(iii) at each window and disconnectedness on each interval of Table 1 up to L . Part B recomputes r_0 , $\omega(r_0)$, the thirteen owners, their hire times, X_1 , and the chain $73 \rightarrow 37 \rightarrow 19 \rightarrow 5$. Primality of numbers below 2^{64} is decided by `sympy.isprime`, which is deterministic in that range. The precise finite search behind Table 2 is as follows. For each admissible $j \geq 1$, in increasing order, test $2jQ - \chi_3(j)\chi_3(Q)$; these candidates increase strictly, so the first prime is $\omega(Q)$. Set $B = \omega(r_0)$, enumerate all owners w of M_{31} with $w \leq (B+1)/2$, and for each test every admissible owner candidate at most B . Record its first prime, or $> B$ if there is none. Taking the minimum of the recorded finite times gives X_1 . These bounded searches verify the numerical input to Theorem 7.1; they do not independently verify connectivity at X^* . The archived script is `analysis/hire_rate/hire_checks.py` at commit `855964e90c4bb3eb92392b1b2ba585be9fe2f0ef` of <https://github.com/hasjack/nm-notebook>, git blob `6ce1ebd1977e6d7c11211b4c4f28bf70e980f6e1`. From the repository root, with assertions enabled, the run used for the statements above was

```
python3 analysis/hire_rate/hire_checks.py 10000000
```

under Python 3.12.7 and SymPy 1.14.0. It exited with status 0 and printed the transcript below. In that transcript, “Theorem 4” is the script’s name for the component-count check, Theorem 3.5(iii).

```
Part A: windows checked = 664577, violations of Theorem 4 = 0
M=      7  r0=     13  [r0, 2r0-2]=[13, 24]  checked to 24: True
M=     17  r0=     67  [r0, 2r0-2]=[67, 132]  checked to 132: True
M=     31  r0=     61  [r0, 2r0-2]=[61, 120]  checked to 120: True
M=    127  r0=    509  [r0, 2r0-2]=[509, 1016]  checked to 1016: True
M=    257  r0=   4111  [r0, 2r0-2]=[4111, 8220]  checked to 8220: True
M=   8191  r0=  16381  [r0, 2r0-2]=[16381, 32760]  checked to 32760: True
M=  65537  r0= 262147  [r0, 2r0-2]=[262147, 524292]  checked to 524292: True
M= 131071  r0=1834993  [r0, 2r0-2]=[1834993, 3669984]  checked to 3669984: True
M= 524287  r0=1048573  [r0, 2r0-2]=[1048573, 2097144]  checked to 2097144: True
Part B: r0 = 98784247763  m0(r0) = {2: 1, 23: 1, 2147483647: 1}  t = 46
hire time of r0: 1382979468683 = 14 * r0 + 1
owners w of M31 with 2w-1 <= 1382979468683: 13
w=   98784247763  door={2: 1, 23: 1, 2147483647: 1}  hire=1382979468683
w=  292057775993  door={2: 3, 17: 1, 2147483647: 1}  hire=None
w=  300647710579  door={2: 2, 5: 1, 7: 1, 2147483647: 1}  hire=None
w=  313532612461  door={2: 1, 73: 1, 2147483647: 1}  hire=627065224921
w=  395136991049  door={2: 3, 23: 1, 2147483647: 1}  hire=None
w=  420906794813  door={2: 2, 7: 2, 2147483647: 1}  hire=841813589627
w=  472446402341  door={2: 2, 5: 1, 11: 1, 2147483647: 1}  hire=None
w=  485331304223  door={2: 1, 113: 1, 2147483647: 1}  hire=None
w=  511101107987  door={2: 1, 7: 1, 17: 1, 2147483647: 1}  hire=None
w=  519691042573  door={2: 1, 11: 2, 2147483647: 1}  hire=None
w=  584115551983  door={2: 4, 17: 1, 2147483647: 1}  hire=None
w=  601295421161  door={2: 3, 5: 1, 7: 1, 2147483647: 1}  hire=None
w=  635655159511  door={2: 3, 37: 1, 2147483647: 1}  hire=None
first live arc into M31 at X1 = 627065224921  (w = 313532612461 , 2w-1 = 627065224921 )
mainland chain from 73: 73 -> 37 -> 19 -> 5
```

Changes from the earlier version. (1) Lemma 3.1 now records $q \leq (p+1)/2$; the previous “ $< (p+1)/2$ ” was false for $m_0(p) = p+1$. (2) The single first-owner window became an octave (Theorem 5.1). (3) The informal “absorbed into the infinite component” argument became Theorem 3.5 and Theorem 6.1. (4) The M_{31} data became an exact statement about all X up to $2M_{61} - 2$, and the bridge owner v is shown not to be the bridge. (5) Windows are explicitly integer-valued, and the equivalence is stated using arbitrarily large cutoffs. (6) The Linnik

bound is an existence statement for chosen bridge primes, with its reference corrected. (7) First attachment α is distinguished from connection to 5, β ; the unsupported density claim is removed. (8) The owner-search signs, exact table cutoffs and candidate count are corrected, and the exclusions of intermediate Mersenne exponents are explicit.

References

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- [4] S. S. Wagstaff, Jr., Divisors of Mersenne numbers, *Math. Comp.* **40** (1983), 385–397.